

# OCC: Software Developer Computing Theory COT512

## Formative Assessment 2 2026



**CTU**  
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## DECLARATION OF AUTHENTICITY

I Grant Samson (FULL NAME) hereby declare that the contents of this assessment COT512 FA2 is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any **AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.**

| Activity | Date |
|----------|------|
|          |      |
|          |      |
|          |      |
|          |      |
|          |      |
|          |      |

Signature: \_\_\_\_\_



Date: 21/08/2026

# Pre-Assessment Agreement

## Assessment Preparation: Preparing the Candidate

|                                                                                                          |                                                                                                                                                                                |              |                 |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|
| Name of Candidate                                                                                        | Grant Samson                                                                                                                                                                   | Date         | 24/08/2026      |
|                                                                                                          |                                                                                                                                                                                | Time         | 09:00           |
| Name of Assessor                                                                                         | MR NB Dube                                                                                                                                                                     | Venue        | Campus          |
| How to prepare the candidate                                                                             | Document Requirements                                                                                                                                                          | Agree (tick) | Action Required |
| Explain to the candidate why you are meeting and the purpose of the assessment.                          | Assessment Policy<br>Assessment process                                                                                                                                        | ✓            |                 |
| Discuss the assessment plan in detail.                                                                   | Assessment strategy                                                                                                                                                            | ✓            |                 |
| Explain assessment process, show assessment instruments to candidate and describe assessment conditions. | Assessment instruments                                                                                                                                                         | ✓            |                 |
| Identify the role-players during assessment.                                                             | Assessors<br>Moderator                                                                                                                                                         | ✓            |                 |
| Describe the evidence required to be declared competent.                                                 | Examples of evidence                                                                                                                                                           | ✓            |                 |
| Explain how evidence will be judged.                                                                     | Mark allocation explained                                                                                                                                                      | ✓            |                 |
| Explain to the candidate how to prepare: Give candidate assessment task description.                     | Assessment task description                                                                                                                                                    | ✓            |                 |
| Confirm with the candidate what he/she should bring to the assessment.                                   | Detailed briefing on exact requirements to be given to candidate in writing                                                                                                    | ✓            |                 |
| Ensure that candidate understands the procedures of all assessment practices.                            | Appeals Policy<br>Appeals procedure<br>Assessment Policy<br>Assessment Procedure<br>Moderation Policy<br>Moderation procedure<br>Verification Policy<br>Verification Procedure | ✓            |                 |
| Ask the candidate if he/she foresees any problems or identify any special needs.                         | List needs                                                                                                                                                                     | ✓            |                 |

## Agreed Assessment Plan

|                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                    |                                                   |
|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------------------------------|
| Candidate's Name:                                                                                                | Grant Samson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                                                   |
| Assessor's Name:                                                                                                 | Mr Nb Dube                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                    |                                                   |
| Module name:                                                                                                     | Computing Theory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                    |                                                   |
| Unit Standard/s:                                                                                                 | <ul style="list-style-type: none"> <li>❖ KM-05-KT01: Introduction to programming languages</li> <li>❖ KM-05-KT02: Programming basics</li> <li>❖ KM-05-KT03: Software applications</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                                                   |
| Type of Assessment<br>i.e. Formative<br>assignment,<br>Formative test,<br>Formative Practical,<br>Summative etc. | Formative Assessment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                    |                                                   |
| Special Assessment<br>Requirements:                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                    |                                                   |
| Event                                                                                                            | Date, time and<br>location                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Resources required | Evidence to be<br>generated                       |
| Assessments due<br>dates.                                                                                        | 28/08/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Assessments        | Completed<br>documentation                        |
| Complete activity<br>upload to Campus<br>Online                                                                  | 28/08/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Study guide        | Completed Portfolio<br>of Evidence                |
| Submit Electronic<br>Portfolio of Evidence<br>on EPOE4ALL                                                        | 28/08/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | CampusOnline       | Acknowledgement of<br>receipt<br>from Facilitator |
| Assessor Roles and Responsibility                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                    |                                                   |
| Roles                                                                                                            | <ul style="list-style-type: none"> <li>❖ Assessor</li> <li>❖ Guide</li> <li>❖ Feedback Agent</li> <li>❖ Reviewer</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                    |                                                   |
| Responsibilities                                                                                                 | <ul style="list-style-type: none"> <li>❖ Consult candidate re-assessment, assessment process and plan.</li> <li>❖ Agree assessment process and plan with candidate.</li> <li>❖ Forward documentation to candidate: plan, guide and assessment instruments.</li> <li>❖ Assess candidate with the use of different instruments.</li> <li>❖ Provide feedback on assessment findings.</li> <li>❖ Support candidate through assessment process.</li> <li>❖ Source feedback from candidate on assessment process.</li> <li>❖ Review assessment process and outcome.</li> <li>❖ Use assessment process as opportunity to transform assessment activities and outcomes.</li> </ul> |                    |                                                   |

| Candidate Roles and Responsibility                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Roles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>❖Learner</li> <li>❖Feedback agent</li> <li>❖Reviewer</li> </ul>                                                                                                                                                                                                                                                                                                                                            |
| Responsibilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>❖Be available for assessment.</li> <li>❖Be actively involved in the consultative process.</li> <li>❖Learn from the assessment process.</li> <li>❖Provide feedback to the assessor in terms of the assessment as learning activity.</li> <li>❖Provide feedback to the assessor on the efficacy of the assessment process.</li> <li>❖Review own role and assessor role in the assessment process.</li> </ul> |
| Assessment Instruments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>❖Electronic Portfolio of Evidence</li> <li>❖Questionnaire</li> <li>❖Report</li> <li>❖Presentation</li> <li>❖Reflexive questions</li> <li>❖Work sample</li> <li>❖Practical's</li> <li>❖Group Activity</li> <li>❖Research activities</li> </ul>                                                                                                                                                              |
| Assessment Process                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <ul style="list-style-type: none"> <li>❖Evaluation of E-POE addressing Essential Embedded Knowledge in unit standards.</li> <li>❖Evaluation of Research Projects and other evidence addressing specific unit standards.</li> <li>❖Consultation: assessment plan and assessment activities and instruments. Pre-assessment moderation and interviews conducted at this stage.</li> <li>❖Observation: feedback on assessment against specific outcomes, critical outcomes in unit standards.</li> <li>❖Feedback: to candidate regarding sufficiency of evidence and possible interview to gain supplementary evidence.</li> <li>❖Feedback to candidate regarding assessment findings as well as review process.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Feedback                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Written feedback to be given to all stakeholders at the end of the assessment process, as well as verbal feedback to the candidate during assessment activities.                                                                                                                                                                                                                                                                                  |
| Recording Process                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Process and findings to be recorded and submitted for record keeping purposes as well as moderation and verification.                                                                                                                                                                                                                                                                                                                             |
| Review Process                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>The review process is the responsibility of the assessor and the candidate.</p> <p>Joint reviewing will take place after feedback has been given to the candidate.</p>                                                                                                                                                                                                                                                                         |
| Right to appeal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | The candidate must be advised of the right to appeal.                                                                                                                                                                                                                                                                                                                                                                                             |

|                    |                                                           |
|--------------------|-----------------------------------------------------------|
| Resources Required | Assignments<br>E-POE<br>Assessments<br>O'Reilly<br>Guides |
|--------------------|-----------------------------------------------------------|

I confirm that:

- ❖ I have been consulted on and have agreed to the training and assessment process as detailed in the assessment guide.
- ❖ I have been advised of my right to appeal against any assessment that is unfair, unreliable, invalid or impracticable.
- ❖ I have read and understood the appeal procedure.
- ❖ I know that assessments may be moderated or verified by an external party.
- ❖ The purpose of the assessment has been clearly explained to me.
- ❖ The criteria have been discussed with me, and I know I will be assessed against these criteria.
- ❖ I know when and where I will be assessed, and I was given fair notice.
- ❖ I know how the assessment will be done, and any other requirements related to the assessment.

Signed: \_\_\_\_\_



Date: 21/08/2026

## QUALIFICATION AND ASSESSMENT DETAILS

|                           |                         |
|---------------------------|-------------------------|
| <b>Programme Title</b>    | OCC: Software Developer |
| <b>Module Title</b>       | Computing Theory        |
| <b>Module Code</b>        | COT512                  |
| <b>NQF Level</b>          | 5                       |
| <b>Assessment</b>         | Formative Assessment 2  |
| <b>Hand Out Date</b>      | 24/08/2026              |
| <b>Hand in Date</b>       | 28/08/2026              |
| <b>Total Marks</b>        | 100                     |
| <b>Developer</b>          | Kobus Mienie            |
| <b>Internal Moderator</b> | Lerato Tsoehlisi        |
| <b>External Moderator</b> |                         |

|                                 |                     |
|---------------------------------|---------------------|
| <b>Student Name and Surname</b> | <b>Grant Samson</b> |
| <b>Student Number</b>           | <b>20261424</b>     |

## INSTRUCTIONS

- Read each question carefully and consider the mark allocation before answering.
- Ensure you answer all questions.
- For final submission, include the Declaration of Authenticity, Pe-Assessment Document, Assessment document, and all the relevant files required to answer the questions in a zip folder.
- Only a .zip will be allowed for zipped folders
- Font style to be used Calibri (Body) 12pt, unless stated otherwise by a question
- Marks will be deducted for missing or incorrectly named documents

## ASSESSMENT OUTCOME

- KM-05-KT01: Introduction to programming languages
- KM-05-KT02: Programming basics
- KM-05-KT03: Software applications

## QUESTION 1

(100)

### Scenario

Movie Stars DVD Rentals is a small but busy video rental store that rents out DVDs and Blu-ray movies to customers in the local community. The store currently manages its records manually using text files, which often leads to errors such as lost rental information, incorrect fines, and difficulty tracking available movies.

To improve efficiency, the company has decided to develop a simple Python-based Movie Rental Management System. The system must allow staff to manage customers, track rented movies, calculate fines for late returns, and store all data in text files for future use.

You have been appointed as a junior developer to design and implement this system.

Consult rubric for mark allocation.

### Project Requirements:

#### 1. Project Structure

Create the following Python files:

- MovieMain.py – Main program (menu-driven application)
- MovieClass.py – Contains the Customer class

Ensure proper file organisation and separation of logic.

#### 2. Customer Class (MovieClass.py)

Create a Customer class with the following requirements:

- Attributes (Private)
- Customer Name (string)
- Customer Email (string)
- Movies Rented (list)
- Fines Owed (float or integer)

Methods

- rentMovie(movie) – Adds a movie to the customer's rented list
- returnMovie(movie) – Removes a movie from the rented list
- payFine(amount) – Reduces the customer's outstanding fines

Ensure proper encapsulation (use private attributes where appropriate).

#### 3. Main Application Logic (MovieMain.py)

The application must run continuously until the user chooses to exit.

Data Loading

- Load movies from Movies.txt into a list or dictionary
- Load customers from Customers.txt (if file exists)
- Create Customer objects for each loaded customer and store them in a list

#### 4. Main Menu System

Display the following menu:

- 1 Rent Movie
- 2 Add Customer
- 3 Return Movie
- 4 View Outstanding Rentals
- 5 View Outstanding Fines
- 6 Pay Fines
- 0 Exit System

#### 5. Functional Requirements

- Option 1: Rent Movie
  - o If no customers exist, prompt to create one
  - o Display available movies
  - o Allow selection of movie
  - o Display customers and allow selection
  - o Decrease available copies of selected movie
  - o Add movie to selected customer's rented list
- Option 2: Add Customer
  - o Prompt for:
    - Customer name
    - Customer email
  - o Create Customer
- Option 3: Return Movie
  - o Display customers and their rented movies
  - o Select customer and movie to return
  - o Increase available copies of movie
  - o Remove movie from customer
  - o Ask if movie is late:
    - If yes, enter number of days late
    - Calculate fine: R5 per day
    - Add fine to customer account
- Option 4: View Outstanding Rentals
  - o Display all customers and movies currently rented
- Option 5: View Outstanding Fines
  - o Display customers with outstanding fines

- Option 6: Pay Fines
  - o Display customers with fines
  - o Select customer
  - o Enter payment amount
  - o Deduct amount from fines owed
- Option 0: Exit System
  - o Save all customer data to Customers.txt
  - o Save all movie data to Movies.txt
  - o Close the application safely

## 6. Programming Requirements

- Use functions to structure the program logically
- Include meaningful comments throughout the code
- Ensure input validation (where applicable)
- Proper use of lists/dictionaries
- Clean and readable code structure

End of Assessment

## MARKING RUBRIC

| Task                                               | Marks<br>Allocated | Marks<br>Awarded |
|----------------------------------------------------|--------------------|------------------|
| Project structure (correct files and setup)        | 10                 |                  |
| Customer class design (attributes + encapsulation) | 10                 |                  |
| Customer methods (rent, return, pay fine logic)    | 10                 |                  |
| File handling (loading movies and customers)       | 10                 |                  |
| Menu system implementation                         | 10                 |                  |
| Rent movie functionality                           | 10                 |                  |
| Return movie + fine calculation                    | 10                 |                  |
| Add customer functionality                         | 5                  |                  |
| View rentals and fines features                    | 5                  |                  |
| Pay fines functionality                            | 5                  |                  |
| Save data to text files on exit                    | 5                  |                  |
| Use of functions and modular design                | 5                  |                  |
| Comments, readability, and coding style            | 5                  |                  |
| <b>Total:</b>                                      | 100                |                  |

### MARK ALLOCATION

| Task       | Description                    | Marks |
|------------|--------------------------------|-------|
| Question 1 | Movie Stars DVD Rentals System | 100   |

### LEARNING OUTCOMES INTEGRATED QUESTIONS CROSS-MAPPING GRID

| LO No.     | Learning Outcome Description          | Question | Marks |
|------------|---------------------------------------|----------|-------|
| KM-05-KT01 | Introduction to programming languages | 1        | 100   |
| KM-05-KT02 | Programming basics                    |          |       |
| KM-05-KT03 | Software applications                 |          |       |
| Total      |                                       |          | 100   |